MUHAMMAD ZAHID	BT401 Genetic Resources & Conservation
MCQs	Fall 2020 (December 2020)
Grand Quiz	Midterm

1. Conservation of plant genetic resource is necessary for ____ ……Both A&B (food security and agrobiodiversity)
2. A concrete plan of action especially designed to achieve concrete goals is called……strategy
3. PGRFA stands for …… plant genetic resource for food and agriculture
4. PGRA stands for …… plant genetic resource for food and agriculture
5. NCCP stands for ……Natural cultural collection of Pakistan
6. IBPGR stand for …… international board of plant genetic resource
7. FGR stands for …… Forest genetic resources
8. AnGR stands for …… Animal genetic resources
9. IUCN stands for …… International Union for Conservation of Nature
10. WCPA stands for …… World Commission on Protected Areas
11. CBD stands for …… Convention on Biological Diversity
12. ABSA is abbreviation of …… access and benefit sharing agreement
13. PGR is abbreviation of …… plant genetic resources
14. GSPC stands for …… Global Strategy for Plant Conservation
15. GEWIS stand for……A genome wide interaction scan
16. Gene bank is the type of …… Ex Situ conservation
17. According to FAQ species of mollusk are ……85000
18. Conservation of plant genetic resource is necessary for……All of them
19. The global strategy for plant conservation GSPC is a program to save the world plant founded in …… 1999
20. The plant that grows spontaneously in self-maintaining populations in natural or semi-natural ecosystems and can exist independently.” are called …… wild plant

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21. The plant resources that grow spontaneously in self-maintaining populations in natural or semi-natural ecosystems and can exist independently." are called …… **wild plant resources**

22. Total area of major pulse/legume crop in Pakistan……..**1.5m hectares**

23. Plant diversity is urgently & effectively conserved is an objective of ……..**GSPC**

24. which disease spread due to consuming mercury poisoned fishes by people …… **MINAMATA disease**

25. _____is an interdisciplinary subfield that aims to understand the dynamics of genes in population to avoid extinction…… **Genetic conservation**

26. Ex-Hepatin an _____is used to control clots in blood, is obtained from Ox lungs…….. **anti-coagulant**

27. Tiger's claw is another name for ……**Coral tree plant**

28. The combination of all the genes present in a given population is called the ……..**Gene pool**

29. _____are plant or population having unique valuable germplasm which are generated…… **Genetic stocks**

30. Genetic material is ……..**functional units of heredity**

31. Increasing genetic diversity interprets daughter chromosomes contain genes from …… **both parents**

32. Resources have to be maintained in such a state that it minimize the risk of loss and allow either its …… **plants genetic resources conservation / Germplasm conservation**

33. Domesticated animals' resources are important due to ____ reasons…….. **All given**

(Provide milk, Provide hair, Provide meat)

34. Crossing over is ___…….. **Exchange of segments between chromosomal section**

35. The activity of keeping animals or plants in order to produce animals or plants is called …….. **Breeding**

36. Genetic resources for food, agriculture and forestry include…….. **Both wild species and domesticated forms**

37. Maintenance of ecosystem is main goal of conserve ------ resources……**Genetic**

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38. _____is the process of protecting an endangered species by placing in an artificial environment…… **Ex situ**

39. _____ lines/ populations developed in breeding programs have narrow genetic base and often contain valuable gene combination…… **Breeding**

40. A phenomenon occurring during prophase I of meiosis is…… **Crossing over**

41. The varieties which were developed by systematic breeding effort. were once commercially cultivated but are no more grown are called…… **obsolete**

42. _____ is used for plant varieties that usually breed through self-pollination…… **Line breeding**

43. The areas of great genetic diversity are protected from human interference are known as…… **Gene Sanctuary**

44. Sum total of all hereditary material that can be used to maintain plant and animal breeding is called……**Germplasm**

45. Scientific name of house cat is…… **Felis catus**

46. The article no says that "The Contracting parties shall develop and maintain appropriate policy and legal measures to promote the sustainable use of plant genetic resources for food and agriculture" …… **Article 6**

47. Animal genetic resources refereed as…… **All**

48. _____ % of the total land of Pakistan is covered by covering vegetables is……**75%**

49. Sharp lowering of population's gene pool because of an environmental or human-caused, change is…… **Bottleneck effect**

50. 'The Global Strategy for Plant Conservation (GSPC) is a program of the UN's Convention on -----diversity founded in 1999…… **Biological**

51. National parks are the examples of _____ conservation…… **In Situ**

52. Genetic stocks can be divided into ___ general groups……**3**

53. Out of total global population of Markhor_____ number is found in Pakistan…… **1500**

54. The term mycorrhiza refers to the role of the _____ in the plant's rhizosphere, its root system…… **Fungus**

55. Scientific name of wild horse is ____……**Equus ferus**

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56._____ conservation means the conservation of ecosystems and natural habitats and the maintenance and recovery of via populations of species in their natural surroundings…… **In situ**

57. The diversity of life on earth is essential to ……..**the survival of humanity**

58._____ could be called as genetic resource…….. **Germplasm**

59. Resources are classified into _____ Types……**3**

60. The term animal genetic resources (AnGR) is used to include…… **all animal species, breeds and strains**

61. What is the consequence of a change in the genetic diversity?..... **Loss of biological diversity**

62. Aquatic genetic resources include all genetic resources living in:…… **water**

63.A species of plant or animal that is in immediate danger of becoming extinct and needs protection to survive is called_____.....**Endangered species**

64. Plant genetic resources are used by farmers and scientists as the raw material for…….. **Breeding new plant varieties and in biotechnology**

65._____ broadly defined as plants or population generated and/or selected for gene…….. **Genetic stocks**

66.GSPC was founded in **1999**

67. Peste des petits ruminants also known as..... **Sheep & goat plague**

68. Woody plants like trees have a _____ genetic diversity than vascular plants like grasses……..**Higher**

69. In ecosystem microorganisms are important for the reason…… **All of these**

(Symbionts, Nitrogen fixers, Involve in Biodegradation)

70._____ are the link from generation to generation of all living matter…….**Genes**

71. The strategy of GSPC seeks to halt the continuing loss of ____ diversity…… **Plant**

72._____ are the areas of terrestrial and coastal ecosystems promoting solution to reconcile the conservation of biodiversity …… **Biosphere reserve**

73. Other species of crops are derived from ……..**Wild crops**

74. Natural and cultural heritage protected permanently by …… **In Situ conservation**

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75. MINAMATA disease was spread due to consuming ___ poisoned fishes by people…….. **Mercury**

76. The special about agricultural genetic resources are …… **To feed the world**

(To feed the world, Conservation, Food security)

77. The transfer of genes from parents to offspring is called …….. **Vertical genetic transfer**

78. The global strategy for plant conservation (GSPC) is …… **useful for plant conservation**

79. Living genetic resources such as seeds or tissues that are maintained for the purpose of animal and plant breeding are called ……..**germplasm**

80. Issues such as predation and hunting can be monitored and managed more easily by _____ conservation……..**Ex Situ**

81. Alarming situation of the earth can be measured by _____species…… **endangered**

82. Direct and indirect impact of human on environment is termed as ……**Anthropogenic effect / Genetic effect**

83. _____ is the process whereby wild plants have been evolved into crop plants through artificial selection…….. **plant domestication**

84. A goat is hoofed mammal in the genus ……..**capra**

85. Bottlenet effect increase the effect of…… **Gene diversity**

86. _____ of the total land of Pakistan is covered by covering vegetables……..**75%**

87. In_____ Phenomenon the original population can give rise to different new populations…… **Founder effect**

88. Madhuca insignis is a species of plant in the family……**Sapotaceae**

89. Causes of migration in animals are ……..**All of them**

(Availability of food, Fast deposition, Photoperiodism, All of them)

90. "superior offspring of quality." are produced in……**Outbreeding**

91. The migration of birds by powerful wind and hurricanes to very long distance……..**Irregular migration**

92. The _____ is the reduction in genetic variation that results when a small subset of a large population is used to establish a new colony……**Founder Effect**

93. Genetic stock can be divided into ----------general groups……..**3**

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94. The frequency of genes in gene pool can be affected by……**all of them (gene flow, mutation, founder effect, inbreeding, natural selection, genetic drift)**

95. The last member of passenger pigeon, named " ____ " and died at age 29 at the Cincinnati Zoo in 1914……**Martha**

96. Gene transfer between different species occur in……**Horizontal gene transfer**

97. The degree of relationship among breeding is less intense than in …. Breeding system……**Line breeding**

98. Genetic drift shows changes in the genome that may be……**All of them (Advantageous, Deleterious, Have no effect)**

99. The species that help to define entire ecosystem is……**keystone species**

100. Transfer of genetic variation from one population to another is termed as……**Gene flow**

101. Dwarf lake iris is a……**plant / Blue flower**

102. The transfer of DNA from one cell to another by a virus……**Transduction**

103. Hazard to animals migration caused by human action or human involvement are called……**anthropogenic hazards**

104. Sympatric speciation is due to……**Reproductive isolation**

105. Sub species of black rhino was declared extinct in……**2011**

106. Northern elephant seals have reduced genetic variation due to……**bottleneck effect**

107. Populations with a lesser number of genes in their gene pool will be……**All given**

108. Salmon move to …… for spawning……**river**

109. Mating of relatives beyond 2nd generation and upto 6th generation. Comes under……**Mild Inbreeding**

110. Scientific name of Marco Polo sheep is……**Ovis ammon**

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111. Migration is… occur for the survival of organism……..**behavioral adaptation**

112. Preservation of biological material in liquid nitrogen is called…….**Cryopreservation**

113. According to IUCN extinction rate of amphibians is……**25-30 % (highest 30%)**

114. According to IUCN _____ specie has highest rate of extinction…… **amphibians**

115. Bubal Hartebeest which had extinct was an……..**animal**

116. The Pitcher's thistle grows for _____ years before its flowers……**4 to 5**

(correct is 5 to 8 years, but in quiz it is not given)

117. Vitrification is solidification due to increased viscosity rather than crystallization technique used in……**Cryopreservation**

118. Specific group of domestic animals with a homogeneous appearance……**breed**

119. Genetic resources are sometimes called……**first resource**

120. Landraces is a……**local variety of domesticated plants**

121. _____is the reduced biological fitness in a given population as a result of inbreeding or breeding of related individuals……**Inbreeding depression**

122. Which is called rosewood……**Dalbergia Sissoo / Sheesham tree**

123. Which one is extinct features of species……..**white rhino**

124. Total remaining snow leopard in Pakistan……**200**

125. Microorganisms produces in……**Antibodies**

126. According to world database on protected area there are over……**210,000 protected areas around the world**

127. Rhino are hunted for……**horns**

128. Natural selection will always result in the---------- that gives an advantage to its Possessor……**selection of allele**

129. Cryopreservation is done in liquid nitrogen……**at -196 degree**

130. Cryopreservation is done over solid carbon dioxide……**at -79 degree**

131. Cryopreservation is done in low temperature & deep freezer……**at -80 degree**

132. Cryopreservation is done in vapor phase nitrogen……**at -150 degree**

133. How many categories of gene flow……**2**

(vertical and horizontal gene transfer)

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134. Breeding type that is not closer to half-sister half-brother and cousins……close breeding / Line breeding

135. The strongest effect of genetic drift on……small population

136. Themes of genetic resources……2

137. Plant genetic resources are the building blocks and fundamental not only in ____but also for the very survival of the species in time and space….crop improvement program

138. Biologist suspect we're living through the --------mass extinction……Sixth major

139. Honey bee pollinates how much plants……250,000 species of plants

140. _____is a pioneer species easily regenerated from seed……Acacia nilotica

141. Sheep was domesticated by humans in……10,000BC

142. Allele frequency affected by___such as mountain…… evolutionary mechanism

(*gene flow *inbreeding* natural selection* founder effect* random genetic drift)

143. Population bottleneck in 1890s due to…… overhunting

144. Reason for crustaceans migration?……for breeding

145. Sterculia khasiana was endemic tree of……Khasi Hills In Meghalaya In India

146. Which involves storage in high and low temperature……cold storage

147. Which of the following is the earliest modern protected area?……yellowstones national park

148. Bees are responsible for pollinations……250,000species

149. Mild form of breeding……line breeding

150. Type of inbreeding……3 (close inbreeding, mild breeding, line breeding)

151. Longest migration was observed in……artic tern bird

152. Cryopreservation of tissue is done at the following temperature …… -196 ^{0}C

153. Calypha belongs to family_____ ……Euphorbiaceae (spurge family)

154. Cause behind extinction of Steller sea cow……hunting

155. Darwin finches is example of……adaptive radiation

156. Result of environment and human change……bottleneck effect

157. Genetic diversity is the variation of individual……amount of genetic information

158. Sigillaria was a tree……spore bearing

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159. The tecopa pupfish was native to the……..**mojava desert**

160. Extinction rate of mammals……**20%**

161. Which of the following approach can be taken to analyze GxE interaction……..**polygenic approach**

162. Northern areas of Pakistan serving as…… **habitat**

163. Population of Himalayan brown bear……**150-200 in Pakistan**

164. Which is used to protect individuals in protected environment……**In situ**

165. Which is used large scale for food and agriculture……**sheep**

(pig, goat, sheep, cattle, buffalo, horse, donkey/Ass, Dromedary, Bactrian camel are used for food and agriculture)

166. Example of founder effect?……**all (haemophilia, dwarfism, infant death)**

167. Which is example of in situ……**all (national park, biosphere, Gene sanctuaries)**

168. Pakistan is the --------- largest producer of kinow oranges……**6th**

169. In 19th century only elephant seals were present……**20**

170. Coral Reef are home to _____ marine animals……**25%**

171. Gene pool types……**3 (Primary GP1, secondary GP2, tertiary GP3)**

172. Crow follow migration……**daily**

173. Dwarf lake iris appearance……**deep blue**

174. Cooling process in which water of tissue become glass rather than crystal……..**vitrification**

175. Cryopreservation is applicable to fishery medical, animal-husbandry……..**all**

176. Cause of extinction……**all (climate change, hunting, over-population)**

177. Yellow stone national park designed……**1872**

178. Sea cow hunted ____ to extinction ……**1768**

179. Extinct plant still cultivated in captivity is……**Cyanea superba**

180. Breeding line have……**narrow genetic base**

181. Species in danger extinction are……**endangered**

182. How many animal species in danger……**20,000 (not confirm)**

183. For which purpose the crustacean can migrate……**Breeding**

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184. Microorganisms are _____ diversity than other living organisms ……**highest**

185. Wildebeest are the part of……**migration**

186. Variation in genetic information of with in of same individual of population……**genetic diversity**

187. Phenylketonuria is the genetic disease caused by……**mutation**

188. _____is a local variety of domesticated plant species……**Landrace**

189. Animal closely related to……**Inbreeding /close breeding**

190. Gene flow blocked by physical barriers results in ……**allopatric speciation**

191. Population bottleneck in 1980s due to……**over hunting**

192. Reef provide habitat to marine life……**25%**

193. Landrace is known as_____……**cultivar**

194. Which of the following is used to protect at suitable environment……**in situ**

195. Trigger of migration……**4**

(local climate, local availability of food, season of the year, for matting reasons)

196. Which is hazard of migration……**all (climate change, drought, food supply)**

197. Natural selection shows ------ change in genome……**positive**

198. Dart-poison of frog yield compound like……**alkaloid**

199. Reef provide habitat to……**marine life**

200. Which of the following is result of reproductive isolation ……**sympatric speciation**

201. Which of the following population has strong genetic drift……**small population**

202. National strategy of PGRFA need to be review if ------- changes……**Country**

203. Main threat to large number of species……**Pollution**

204. If the advantageous gene is removed then the effect is called……**genetic drift**

205. Microorganisms are……**more diverse**

206. Together with the components which fulfill agriecological functions genetic reason are grouped under the concept……**Agrobiodiversity**

207. PGR (plant genetic resources) are components of _____which sustain the humankind……**biodiversity**

208. The degree of relationship is not closer than half-brother or half-sister matings or

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cousin matings…… **line breeding**

209. keystone species ……..**Grizzle bear**

210. Cryopreservation occur….**All (liquid nitrogen, freezing temperature, vapor nitrogen)**

211. Derived genetic diversity except……**inbreeding**

212. ……. Cannot permeate the cell membrane such as……**sugar**

213. ---------- rare treat for eye…….. **Hart's Tongue Fern**

214. The introduction of genetic material by interbreeding from one population of species to another is……**gene flow**

215. An individual can be protected in controlled environment is……**Ex situ conservation**

216. Twins develop from one zygote which splits and forms two embryos are called…….. **Monozygotic**

217. Genes can be transferred sexually or asexually……**vertical transfer**

218. Which one is the reason of bottleneck effect by……..**all of the above**

(mutation, gene flow, inbreeding)

219. Which one of the following is the example artificial barrier…….. **All**

(the great china wall, dams, barrages)

220. Genetic resource are heritable characteristics … of real potential benefit… **both a & b**

[(a) plant, (b) animal, (c) none, (d) both a & b]

221. A membrane is ……**Both a & b**

[(a) permitting, (b) non membrane permitting, (c) a & b, (d) None]

222. Migration involves following type of hazards, ……**2 type**

(natural hazards, anthropogenic hazards to migrants)

223. Trigger of migration……..**all**

(local climate, local availability of food, season of the year, for matting reasons)

224. GEWIS stands for …….. **A genome wide interaction scan**

225. Monach butterflies show migration to escape……**Cold weather**

226. The set of all genes, or genetic information, in any population is called……**Gene pool**

227. Migration of animals from resting site to feeding area is known as……**Daily migration**

228. Allopatric speciation is speciation that results when a population is separated by

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a……**Physical barriers**

(Physical barrier, Population, Natural barrier)

229. A-species may become functionally extinct which are unable to reproduce due to…… **All of them**

(Poor health, Age, lack of individuals)

230. An agent of substance that can bring about a permanent alteration to the physical composition of a DNA gene is…… **Mutagen**

231. Which of the following activity affects the Genetic diversity?..... **Anthropogenic**

232. Man made activities increases the number of changes occurring to individual and populations. Such pressure is exerted by…… **Artificial selection**

233. Breeding for disease resistance requires……**All given**

(Good source of resistance, Disease test, Planned hybridization)

234. Microorganisms are important in Agriculture because ..... **All of the above**

235. The severe reduction in the population size is…… **Bottlenet effect**

236. Random changes in gene frequency especially in small populations when leading to preservation or extinction of particular genes is…… **Genetic drift**

237. The allele frequency in a population's gene pool can be affected by……**All given**

(Mutations, Gene flow, Inter breeding)

238. Preserving germplasm in frozen state is…… **Cryopreservation**

239. Which is a chemical substance produced by microorganisms that can kill or inhibit the growth of other microorganisms…… **Antibiotic**

240. Nearly _____ species of plants and animals are at a high risk of extinction……**20,000**

241. When two different genotypes respond to environmental variation in different ways. It is known as..... **Gene-Environment interaction**

242. Transfer of genetic material from one species to another is called…… **horizontal gene transfer**

243. A specie of plant or animal that is no longer living is..... **Extinct Specie**

244. Genetic drift is change in the allele frequency of a population due to… **Random chance**

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245. Mating with non-relatives is termed as…… **Out breeding**

246. Animals migrate due to reasons…….. **All given**

247. In ___ type of migration only male members move a short distance…….. **Moult**

248. A specie that is likely to become endangered if it is not protected is.**Threatened Specie**

249. Population at with small gene pools are in danger of acquiring……..**All given**

250. Genetic drift is largely influenced by…….. **Population size**

251. The effect of genetic diversity increase as the population size ____ ……**Decreases**

252. Preserving germplasm in frozen state is ____.....**Cryopreservation**

253. _____molecule can freely diffuse through the cell membrane…… **All given**

254. Pure line breed refers to _____..... **Homozygous only**

255. The transfer of DNA from one cell to another through cell-cell contact often involving plasmid is called……**conjugation**

256. Out breeding increases the ……**Heretozygosity**

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